TT-V-71H
January 31, 1979
SUPERSEDING
Fed. Spec. TT-V-71G
February 7, 1975

FEDERAL SPECIFICATION

VARNISH, INTERIOR, FLOOR AND TRIM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers an oleoresinous varnish intended for general interior use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gauge and Lighter.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 8010

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American Society for Testing and Materials (ASTM) Standards:

D 154 - Varnishes, Testing.
D 1296 - Odor of Volatile Solvents and Diluents.
D 1544 - Color of Transparent Liquid (Gardner Color Scale).
D 1545 - Viscosity of Transparent Liquid by Bubble Time Method.
D 1642 - Elasticity or Toughness of Varnish.
D 1643 - Gas Checking and Draft Test of Varnish Films.
D 2369 - Volatile Content of Paints.
D 3272 - Vacuum Distillation of Solvents from Solvent for Analysis.
D 3278 - Flash Point of Liquids by Setaflash Closed Tester.
D 3335 - Low Concentrations of Lead in Paint by Atomic Absorption Spectroscopy.

(Application for copies should be addressed to the American Society for Testing and Material, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Solvent. The solvent, when tested as specified in 4.3.1, shall conform to the following requirements by volume:

(a) The total of solvents with olefinic or cyclo olefinic unsaturation shall not exceed 5 percent.

(b) The total of aromatic compounds with 8 or more carbon atoms in the molecule, except ethylbenzene, methyl benzoate, and phenyl acetate, shall not exceed 8 percent.

(c) The total of ethylbenzene, toluene, and branched-chain ketones shall not exceed 20 percent.

(d) A solvent which may be classified into more than one of the above groups shall be considered a member of the group having the lowest allowable concentration.

(e) The total of (a), (b), and (c) shall not exceed 20 percent.

(f) Halogenated solvents shall not be present.

3.2 Quantitative requirements. The quantitative requirement shall be as specified in table I.

TABLE I. Quantitative requirements

Characteristics	Requirements minimum	Requirements maximum
Drying time		
Set-to-touch, hours	0.5	2
Dry hard, hours	-----	7
Free from after-tack, hours	-----	24
Flash point, °C	26	-----
Lead, percent of lead (Pb) metal in total nonvolatile	-----	0.06
Nonvolatile content, percent by weight	45	-----
Toughness, percent	25	-----
Viscosity, Gardner-Holt	C	H

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3.3 Qualitative requirements.

3.3.1 Appearance. When tested as specified in 4.3.6, the varnish shall be clear, homogeneous, and free from insoluble matter.

3.3.2 Brushing properties and appearance of dried film. When tested as specified in 4.3.4, the varnish shall be a freely-working product with acceptable flowing and leveling properties. The dried film shall be clear, smooth, and glossy, and free from streaks, blisters, and lap marks.

3.3.3 Color. When tested as specified in table II, the varnish shall be no darker in color than the No. 11 tube of the Gardner color scale.

3.3.4 Condition in container. When tested as in 4.3.3, a freshly-opened container shall show no skinning, livering, thickening, and gelling.

3.3.5 Draft resistance. The dried film shall show no pitting, wrinkling, and crow's-footing when subjected to the draft test as specified in Table II.

3.3.6 Gas resistance. The dried film shall show no dulling, frosting, and crow's-footing when subjected to the gas test as specified in Table II.

3.3.7 Odor. The odor shall be characteristic of the volatiles permitted when tested as specified in Table II.

3.3.8 Skinning. The varnish shall show no skinning after 48 hours in a tightly-closed half-filled container, when examined and tested as specified in Table II.

3.3.9 Water resistance. The dried film shall show no whitening and dulling when tested as specified in 4.3.7.

3.4 Quantities. The varnish shall be furnished in 1-pint, 1-quart, 1-gallon, 5-gallon, and 55-gallon quantities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Sampling and Inspection.

4.2.1 Lot. The varnish shall be assembled into lots as specified in MIL-STD-105. In MIL-STD-105, the words "essentially the same conditions" shall be interpreted to mean as manufacturer's batch, which is defined as the end product of all raw materials mixed, blended, or processed in a single operation.

4.2.2 Examination for packaging, packing, and marking. An examination shall be made to determine that packaging, packing, and marking of the end item complies with the applicable requirements of section 5. The sample unit shall be one complete shipping container. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of percent defective.

4.2.3 Sampling of the end item for tests. For the purposes of sampling, the lot shall be expressed in units of gallons. Samples from lots shall be taken in accordance with MIL-STD-105 using inspection level S-2 and an acceptable quality level (AQL) of 2.5.

4.3 Test methods. All tests shall be conducted in accordance with the methods specified in Table II to determine compliance with the requirements of section 3. Unless otherwise specified, standard testing conditions are 23° ± 1°C and a relative humidity of 50 ± 5 percent. All test reports shall contain the individual values utilized in expressing the final result. Failure to pass any test, or noncompliance with any requirement shall be cause for rejection of the sample.

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TABLE II. Index of tests

Test	Requirement Reference	Test Method ASTM Standard	Paragraph Reference
Appearance	3.3.1	-----	4.3.3
Brushing Properties	3.3.2	-----	4.3.4
Color	3.3.3	D 1544	-----
Condition in container	3.3.4	-----	4.3.5
Draft resistance	3.3.5	D 1643	-----
Drying time	Table I	-----	4.3.6
Flash point	Table I	D 3298	-----
Gas resistance	3.3.6	D 1643 Method A	-----
Lead content	Table I	-----	4.3.2
Nonvolatile content	Table I	D 2369	-----
Odor	3.3.7	D 1296	-----
Skinning	3.3.8	D 154	-----
Solvent analysis	3.1	D 3272, D 2267	4.3.1
Toughness	Table I	D 1642 Method A	-----
Viscosity	Table I	D 1545	-----
Water resistance	3.3.9	-----	4.3.7

4.3.1 Solvent analysis. Analyze the solvent by gas chromatography. The solvent shall be separated from the varnish as specified in ASTM D 3272 and then injected into the gas chromatograph, or the varnish shall be injected directly using equipment specifically designed for that purpose. The procedure specified in ASTM D 260, and any apparatus, operating conditions, columns, and options permitted therein shall be used for the chromatographic analysis. The accuracy of the analysis shall be 0.25 percent absolute by weight or less for each component and the reproducibility shall be 0.25 percent absolute by weight over three or more runs. All peaks 0.5 percent of the sample or greater shall be identified and quantified. Convert percent by weight to percent by volume, and evaluate the results for compliance with 3.1.

4.3.2 Lead content. The method described in 4.3.2.1 or the method described in 4.3.2.2 shall be used. The X-ray fluorescence spectrometry method described in 4.3.2.1 shall be the final determinant of compliance in all cases.

4.3.2.1 Analysis for lead by X-ray fluorescence spectrometry. Lead content shall be determined using an X-ray fluorescence spectrometer capable of determining lead content at a minimum level of 0.03 percent by weight. The molybdenum X-ray source shall be operated at 60Kv and 45Ma; a lithium fluoride crystal cut along the 200 planes shall be used to disperse fluorescent X-rays; the detector collimeter shall be set at fine; and flow and scintillation counters shall be used as detectors. Pulse height selection shall be used in all measurements, and the counting time shall be 100 seconds. Place the sample disc in the sample holder, and measure the count rates of lead, lead background, and the molybdenum Rayleigh-scattered background from the X-ray tube at the following angles:

Analytical Line	Angle	Intensity
Mo	20.33	I_{Mo}
Background I	33.50	I'
Pb	33.93	I_{Pb}
Background II	34.50	I"

4.3.2.1.1 Calculation. Determine the ration R of the lead and molybdenum lines as follows:

$$R = \frac{2I_{Pb} - I' - I"}{2I_{Mo}}$$

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4.3.2.1.2 Procedure. Duplicate drawdowns of the well-mixed varnish shall be made on the sealed chart, using a mechanical applicator plate and a film applicator with a 150 um gap. The drawdowns shall be at least 250 mm long, and shall be allowed to air-dry for 24 hours at standard condition. Discs shall then be cut from each drawdown to fit the sample holder of an X-ray fluorescence spectrometer. Standard samples of the same type of paint containing known amounts of lead shall be prepared and measured in the same way, and the R values shall be plotted against the lead concentration to obtain a working curve. By use of the working curve, the value of R obtained for the test sample shall be converted to the lead concentration in percent by weight, and the result shall be evaluated for compliance with the requirement in table II.

4.3.2.2 Analysis for lead by atomic absorption spectrometry. ASTM Method D 3335 shall be used to determine the concentration of lead in percent by weight, and the result shall be evaluated for compliance with the requirement in table I.

4.3.3 Appearance. Fill two test tubes to within 1 inch of the top with thoroughly mixed sample, and stopper the tubes with clean corks. Examine the contents by light, and evaluate for compliance with the requirements of 3.3.3. Fill one of the tubes at a small angle from the horizontal so that the air bubble will move slowly and permit observation in the thin moving film of any fine particle that might otherwise escape notice. Let the tube stand undisturbed for 24 hours and examine for sediment for compliance with the requirement of 3.3.3.

4.3.4 Brushing properties and appearance of dried film. The varnish shall be applied to cold-rolled steel and aluminum panels at a spreading rate of 500 square feet per gallon. It shall be applied in sections, first brushing the coating using back and forth strokes, then leveling the material by brushing at 90° to the first strokes. The brushing properties shall be evaluated for compliance with the requirements in 3.3.4. The varnish shall be allowed to dry and the dried film shall be evaluated for compliance with requirements in 3.3.4.

4.3.5 Condition in container. Examination shall be made with the use of a spatula or paddle and by transferring the material to clear glass container. Evaluate the package condition of the varnish for compliance with 3.3.1.

4.3.6 Drying time. In a well-ventilated room free from drafts, dust, and direct sunlight, place in horizontal position glass panels coated with the sample at a height such that, when the thumb is placed on the film, the arm of the operation is in a vertical straight line from wrist to shoulder. The varnish film shall be 25 $\pm$ 3 um thick. Touch the varnish film lightly by the finger tip, and the film shall be considered set-to-touch when none of it adheres to the finger. With the end of the thumb resting on the test film and the forefinger supporting the test panel, exert a maximum downward pressure of the thumb on the film. Lightly polish the contacted area with a soft cloth. The film shall be considered dry-hard when any mark left by the thumb is completely removed by the polishing operation. Lay a 2- by 3- inch piece of paper on the film and place a cylinder exerting a pressure of 2 pounds per square inch on the paper for 5 seconds. The film shall be considered free from after-tack if, when inverted, the paper drops off the test film within 10 seconds. Evaluate the results for compliance with the requirements of 3.2, table I.

4.3.7 Water resistance. Two panels shall be prepared as specified in ASTM D 1643, Method A, and shall be dried for 48 hours. One of the panels shall be placed in a beaker containing approximately 65 ml of distilled water at room temperature. The end of the panel which was uppermost during the drying period shall be immersed. The panel shall be left immersed in the water for 72 hours. After removal from the water, the panel shall be wiped with a pad of cheese-cloth or cotton to remove excess water, and allowed to dry for 2 hours, and the varnish shall be examined to determine compliance with the requirement of 3.3.9. The second panel shall be tested as above, except that the distilled water shall be boiling and the immersion time shall be 15 minutes.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The 1-pint, 1-quart, and 1-gallon quantities of varnish shall be packaged in metal cans conforming to PPP-C-96, type V, class 2. Exterior Plan B coating shall be required.

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5.1.2 Commercial. The 1-pint, 1-quart, and 1-gallon quantities of varnish shall be packaged in metal cans in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Forty-eight 1-pint cans, or twelve 1-quart cans, or four 1-gallon cans of varnish, packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, grades V3c, V3s, or V2s. The boxes shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Alternatively, wirebound, cleated plywood, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The barrier material shall be sealed at the edges with waterproof tape or adhesive. The 5-gallon and 55-gallon quantities of varnish shall be furnished in metal cans and metal drums conforming to Item 260 of the National Motor Freight Classification and Rule 40 of the Uniform Freight Classification.

5.2.2 Commercial. The varnish shall be packed in a manner that will assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with the National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The product covered by this specification is an interior varnish to be used for floor and trim.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of container required. (see 3.4).
(c) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).
(d) Marking required (see 5.3).

MILITARY INTEREST:

DOD coordination has been waived.

Preparing Activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

Interior-BOR
VA-DMS
Commerce-NBS
HEW-NIH
SBA

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.

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